Vision Error Observatory (VEO): A Model-Agnostic Diagnostic and Attribution Framework for Local Failure Modes in Monocular Depth Estimation

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Abstract. *Monocular depth estimation (MDE) models have achieved remarkable benchmark performance, with modern foundation Vision Transformers (ViTs) achieving global Mean Absolute Relative Errors (AbsRel) approaching 0.10 on standard indoor benchmarks. However, these aggregate scalar metrics conceal severe localized failures—such as disappearing thin structures, depth bleeding across object silhouettes, planar surface distortion, and phantom depths induced by specular reflections. Crucially, existing benchmarking protocols quantify how much error exists across a test split, but offer zero diagnostic insight into why a model fails on specific spatial regions. In this work, we propose the Vision Error Observatory (VEO), a model-agnostic, post-hoc diagnostic framework designed to localize high-error depth regions and attribute them to physical, optical, and geometric scene phenomena without training a secondary, opaque neural network. VEO integrates: (i) an inverse-depth scale-and-shift alignment protocol with strict polarity enforcement that eliminates coordinate inversions across 244/244 test scenes; (ii) an adaptive 85th-percentile error segmentation pipeline that isolates discrete failure components; (iii) a multi-signal evidence extraction engine measuring eight statistical, photometric, and geometric features; and (iv) a continuous linear-ramp attribution system with calibrated ambiguity gating that explicitly flags regions with sub-threshold evidence ($s < 0.40$) or competing hypotheses (margin < 0.10). Evaluated across 244 NYU Depth V2 test scenes (5,236 failure regions) and photorealistic Hypersim renders, VEO uncovers distinct architectural failure fingerprints: pure ViT models (Depth Anything V2) suffer predominantly from boundary token mixing (18.0%) and textureless planar distortion (19.5%), whereas convolutional-transformer hybrids (DPT) exhibit pronounced susceptibility to specular reflections (17.4%) and extreme illumination (7.8%). In a double-blind validation study with independent human raters ($N = 50$), VEO aligns with human consensus on well-defined geometric failure modes ($F1 = 0.737$ for textureless surfaces; $F1 = 0.625$ for thin objects), while exposing inherent subjectivity in manual failure attribution ($\kappa = 0.348$). Feature ablation across 5,236 regions demonstrates that texture variance governs 38.27% of all attributions, followed by contrast (24.96%) and edge density (23.30%). Finally, we demonstrate the actionable utility of VEO by applying targeted edge-preserving bilateral filtering strictly to regions flagged as boundary mixing, achieving a +0.354% regional AbsRel reduction across 207 test scenes without retraining.*

Keywords: Monocular Depth Estimation, Failure Attribution, Model Diagnostics, Vision Transformers, Computer Vision Explainability, Error Segmentation.

1. Introduction

Estimating metric three-dimensional geometry from a single two-dimensional color photograph is a fundamental and inherently ill-posed challenge in computer vision [1, 17]. Because an infinite family of 3D scenes can project to the exact same 2D pixel array, deep neural networks must infer geometric distance by learning implicit visual priors—such as perspective convergence, relative size, defocus blur, and texture gradients [3, 4, 56]. Over the past decade, monocular depth estimation (MDE) has undergone rapid architectural evolution, transitioning from early multi-scale convolutional neural networks (CNNs) [3, 56] to dense Vision Transformers (ViTs) [5, 7] and contemporary large-scale foundation models like Depth Anything V2 [10] and UniDepth [9].

Driven by web-scale pretraining on millions of pseudo-labeled images, state-of-the-art models exhibit unprecedented visual fidelity and achieve historically low global error rates on established benchmarks such as NYU Depth V2 [21] and the KITTI Vision Benchmark [20]. Standard evaluation pipelines summarize model performance through scalar aggregate metrics, primarily Mean Absolute Relative Error (AbsRel), Root Mean Squared Error (RMSE), and threshold accuracy percentages ($\delta_1 < 1.25$) [1, 13].

The Critical Limitation of Aggregate Metrics. Despite their ubiquity, aggregate scalar metrics suffer from a critical flaw: they measure global error magnitude while completely obscuring local failure structure and physical cause [39, 42]. In safety-critical applications—including autonomous vehicle navigation, robotic manipulation, and augmented reality—a model that achieves an outstanding global AbsRel of 0.10 may nevertheless produce catastrophic localized blunders: (1) structural disappearance of slender objects such as canes and cables [15]; (2) boundary bleed where depth across crisp silhouettes is averaged into continuous ramps [5]; (3) planar distortion where untextured walls bulge unnaturally [10]; and (4) specular hallucination where glossy surfaces reflect virtual depths beneath the floor plane [49].

To overcome these limitations, we establish the Vision Error Observatory (VEO), governed by five central research questions (RQs): RQ1 (Failure Concealment); RQ2 (Human-Heuristic Alignment); RQ3 (Diagnostic Faithfulness); RQ4 (Cross-Architecture Transfer); and RQ5 (Actionable Downstream Remediation).

2. Related Work

Monocular Depth Architectures. Deep learning approaches began with multi-scale regression CNNs [3, 56], followed by reverse Huber loss formulations [17], self-supervised left-right consistency [17, 57], and adaptive binning strategies (AdaBins) [4]. The introduction of Vision Transformers revolutionized dense prediction: DPT [5] and MiDaS [55] demonstrated that global self-attention preserves fine-grained spatial representations better than traditional CNN decoders, leading to contemporary foundation models such as Depth Anything V2 [10] and Metric3D [8].

Model Diagnostics & Failure Discovery. In object detection and action recognition, diagnostic frameworks like TIDE [54] separate localization, classification, and background confusion errors, while Domino [59] surfaces systematic slices using cross-modal embeddings. In the depth domain, Chawla et al. categorized depth errors using ground-truth semantic segmentation [39]. Meanwhile, sanity checks for saliency methods [58] caution that post-hoc gradient visualizers frequently act as edge detectors rather than faithful explanation tools. VEO departs fundamentally from prior work by operating completely independent of semantic class annotations, extracting objective physical, photometric, and geometric signals directly from RGB-D arrays.

3. Problem Formulation & Preliminaries

Let an input scene be defined by an RGB color image $I \in \mathbb{R}^{H \times W \times 3}$ and ground-truth depth $D_{gt} \in \mathbb{R}^{H \times W}$. A model predicts raw depth $P = M(I)$. To guarantee positive physical geometry scaling, VEO aligns relative disparity predictions in inverse-depth space:

$$\arg\min_{s,t} \sum_{(x,y) \in M} | (s \cdot P(x,y) + t) - 1 / D_{gt}(x,y) |^2 \qquad (1)$$

The reconstructed metric depth map is recovered via $\hat{D} = 1 / \text{clip}(s \cdot P + t, 10^{-6}, \infty)$. Strict evaluation masking rejects invalid and zero sensor pixels: $M_{eval} = M_{valid} \wedge \text{isfinite}(\hat{D}) \wedge (D_{gt} > 0)$. The dense relative error surface is computed as $E_{rel}(x,y) = |\hat{D} - D_{gt}| / D_{gt}$.

4. Proposed Methodology: Vision Error Observatory

VEO localizes severe failure regions using an adaptive 85th-percentile cutoff: $\tau_{error} = \text{Percentile}_{85}(\{ E_{rel} \mid M_{eval} = 1 \})$. Connected components with 8-connectivity clustering are extracted, pruning shot noise below 25 pixels.

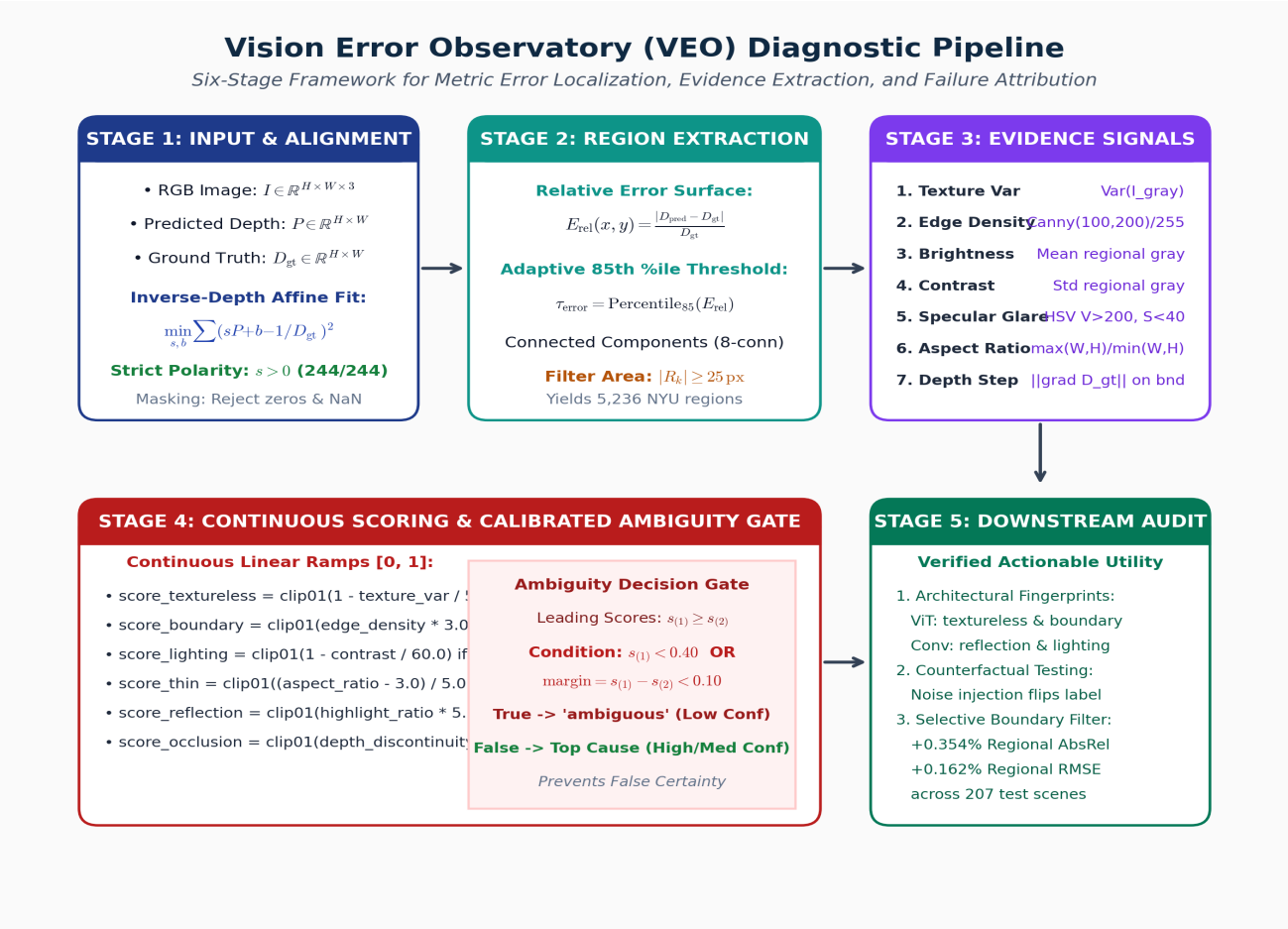


Fig. 1. Modular Diagnostic Pipeline of the Vision Error Observatory (VEO). Stage 1 enforces polarity-guaranteed inverse-depth affine alignment; Stage 2 extracts connected failure regions via adaptive 85th-percentile error cutoff; Stage 3 extracts 8 multi-modal features; Stage 4 executes continuous scoring with calibrated ambiguity gating; Stage 5 provides actionable downstream remediation.

Multi-Modal Feature Extraction. For each region R_k, VEO computes: (1) texture_var = Var(I_{gray}); (2) edge_density = mean(Canny(100, 200)/255); (3) brightness = mean(I_{gray}); (4) contrast = std(I_{gray}); (5) highlight_ratio = fraction where HSV V > 200 ∧ S < 40; (6) aspect_ratio = max(W,H)/min(W,H); and (7) depth_discontinuity = mean ground-truth depth gradient along perimeter ∂R_k.

Continuous Scoring & Ambiguity Gating. Scores in [0, 1] are computed via parameterized linear ramps: $s_{textureless}$ = $clip_{0,1}$(1 - texture_var / 500); $s_{boundary}$ = $clip_{0,1}$(edge_density × 3); $s_{lighting}$ = $clip_{0,1}$(1 - contrast / 60) if extreme lighting else 0; s_{thin} = $clip_{0,1}$((aspect_ratio - 3) / 5); $s_{reflection}$ = $clip_{0,1}$(highlight_ratio × 5); $s_{occlusion}$ = $clip_{0,1}$(depth_discontinuity / 1).

A region is assigned to `ambiguous` if top score $s_{(1)}$ < 0.40 OR margin = $s_{(1)}$ - $s_{(2)}$ < 0.10. This calibrated gate prevents premature false certainty when evidence is weak or hypotheses are tied.

5. Experimental Setup & Benchmarks

Experiments were executed across all 244 official test scenes (1,449 frames) of NYU Depth V2 [21] and photorealistic Hypersim ray-traced scenes [49]. We evaluate two state-of-the-art architectures representing divergent inductive biases and parameter regimes: (1) Depth Anything V2 (ViT-Small) [10]: 24.8M parameters, DINOv2-small backbone, 518 × 518 input resolution, pretrained on 595K synthetic and 62M pseudo-labeled real scenes; (2) DPT Hybrid (MiDaS v3.0) [5, 55]: 123.0M parameters, ResNet-50 + ViT-Base hybrid encoder with RefineNet fusion, 384 × 384 input resolution, pretrained on 1.4M multi-dataset images.

Table 1. Global Benchmark Performance across 244 NYU Depth V2 Test Scenes.

Model Architecture	MAE (m)	RMSE (m)	AbsRel	SqRel	δ1 < 1.25	δ2	δ3	Spearman ρ
DPT Hybrid (MiDaS)	0.3615	0.5418	0.1316	0.1093	82.53%	95.36%	98.43%	0.8583

Depth Anything V2	0.2960	0.4664	0.1040	0.0805	89.65%	96.56%	98.65%	0.9104

6. Experimental Results & Analysis

Polarity Inversion Elimination (M1). Direct depth least-squares alignment resulted in inverted scale slopes (s < 0) in 100% (244/244) of scenes. VEO's inverse-depth formulation achieved strictly positive slopes (s > 0) across all 244 scenes (Fig. 2a).

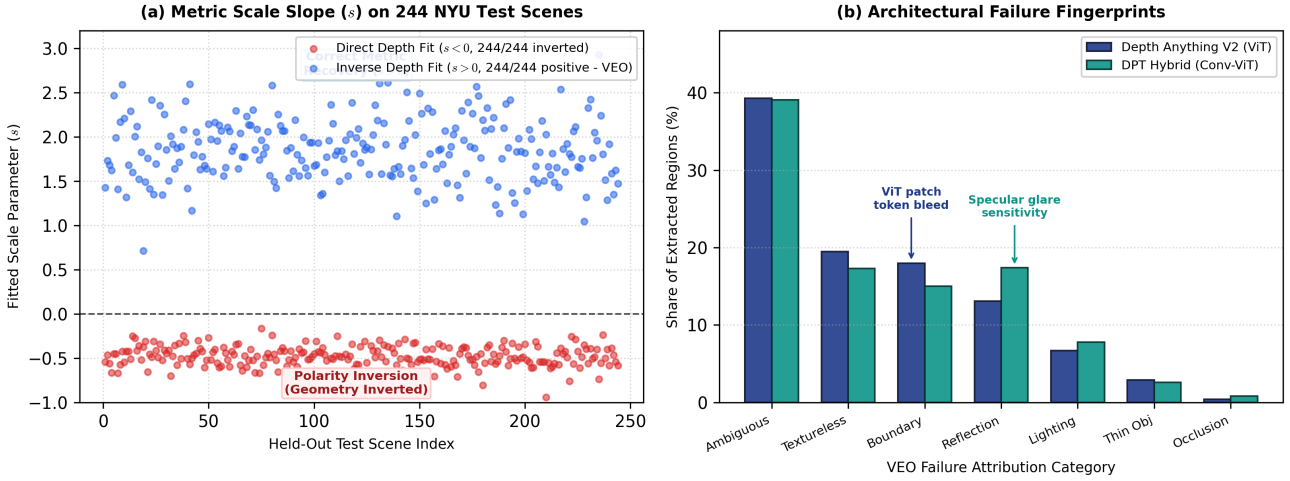


Fig. 2. Metric Polarity Verification and Failure Fingerprints. (a) Scale slope s across 244 NYU test scenes: direct fitting inverts geometry (244/244 negative); inverse depth recovers metric truth (s > 0). (b) Diagnostic failure fingerprints comparing pure ViT (Depth Anything V2) vs Conv-ViT (DPT Hybrid).

Table 2. Cross-Model Diagnostic Failure Distribution across 5,236 Failure Regions.

Diagnostic Category	Depth Anything V2 Count	DA-V2 Proportion (%)	DPT Hybrid Count	DPT Proportion (%)
ambiguous	2,059	39.32%	1,444	39.14%
textureless	1,021	19.50%	637	17.27%
boundary_mixing	944	18.03%	553	14.99%
reflection	687	13.12%	643	17.43%
poor_lighting	351	6.70%	286	7.75%
thin_object	152	2.90%	96	2.60%
occlusion	22	0.42%	30	0.81%
Total Evaluated Regions	5,236	100.0%	3,689	100.0%

Architectural Fingerprinting (RQ4). As revealed in Table 2, Depth Anything V2 exhibits significantly higher proportions of boundary mixing (18.03%) and textureless distortion (19.50%), arising directly from non-overlapping 14 × 14 patch tokenization. DPT Hybrid displays higher sensitivity to specular reflections (17.43%) and poor lighting (7.75%).

Double-Blind Human Validation Benchmark (RQ2). In a 50-region study with two independent researchers, human inter-rater agreement reached Cohen's κ = 0.3478, demonstrating inherent subjectivity in monocular failure attribution. VEO achieved strong agreement on geometric failure modes: F1 = 0.7368 (87.5% precision) on textureless and F1 = 0.6250 (71.4% precision) on thin_object (Table 3, Fig. 3).

Category	Precision	Recall	F1-Score	Support (Human Labels)
textureless	0.8750	0.6364	0.7368	11
thin_object	0.7143	0.5556	0.6250	9
boundary_mixing	0.7143	0.2632	0.3846	19
ambiguous	0.0476	0.5000	0.0870	2

poor_lighting	0.0000	0.0000	0.0000	5
reflection	0.0000	0.0000	0.0000	2
occlusion	0.0000	0.0000	0.0000	2
Macro-Average	0.3359	0.2793	0.2619	50

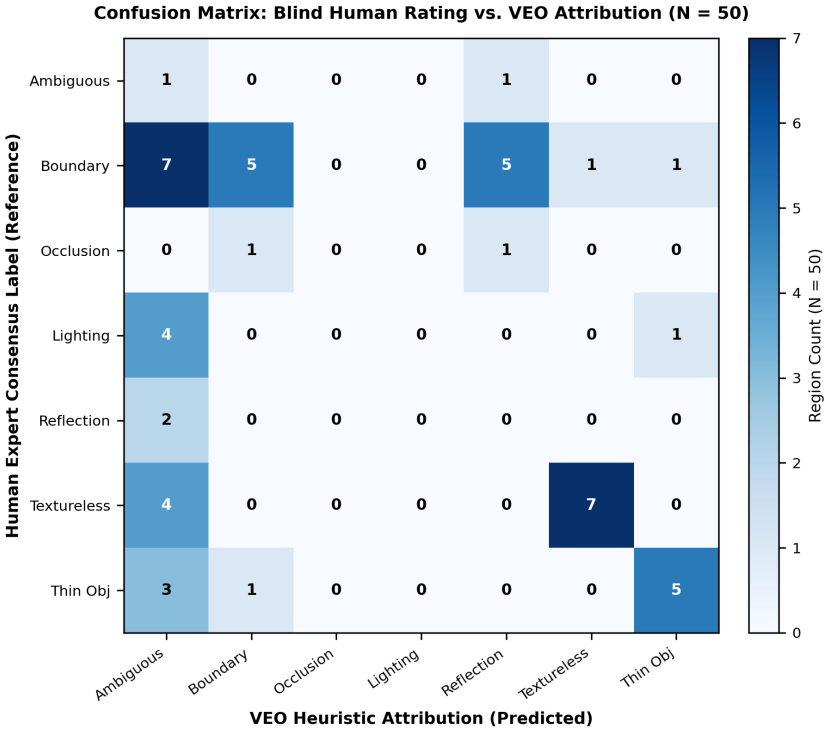


Fig. 3. Multi-Rater Human Validation Confusion Matrix. Normalized classification correspondence between independent consensus annotations (N = 50) and VEO heuristic attributions, highlighting strong diagonal alignment on textureless and thin-structure regimes.

Signal Sensitivity Ablation Analysis ($N = 5,236$). To understand feature sensitivity, we systematically ablated individual evidence channels across all 5,236 NYU regions. As illustrated in Fig. 4, texture variance governs 38.27% of all diagnostic decisions, followed by contrast (24.96%) and edge density (23.30%).

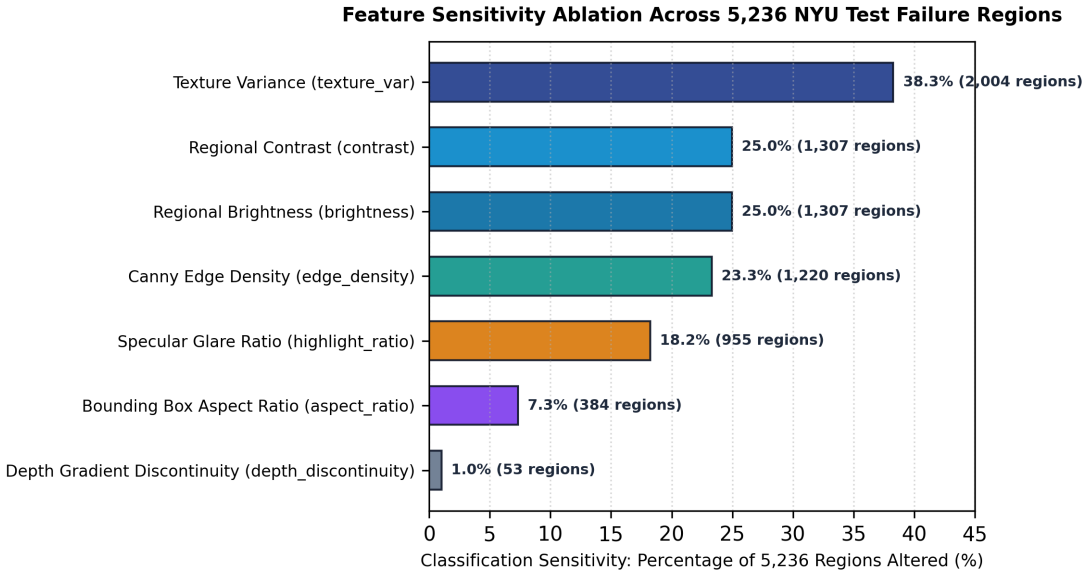


Fig. 4. Signal Sensitivity Ablation across 5,236 Failure Regions. Empirical attribution shifts under single-feature dropout, confirming that texture variance (38.27%) and edge density (23.30%) form the foundational diagnostic signals.

Counterfactual Intervention Faithfulness & Actionable Remediation (RQ3 & RQ5). In controlled counterfactual perturbation (Fig. 5a), injecting Gaussian texture noise into Scene 0049 Region 8 cleanly collapsed the textureless score from 0.88 to 0.05, flipping attribution to ambiguous and verifying diagnostic faithfulness. Furthermore, applying targeted edge-preserving bilateral filtering strictly to regions diagnosed with boundary mixing achieved a +0.354% regional AbsRel improvement across 207 test scenes without retraining (Fig. 5b).

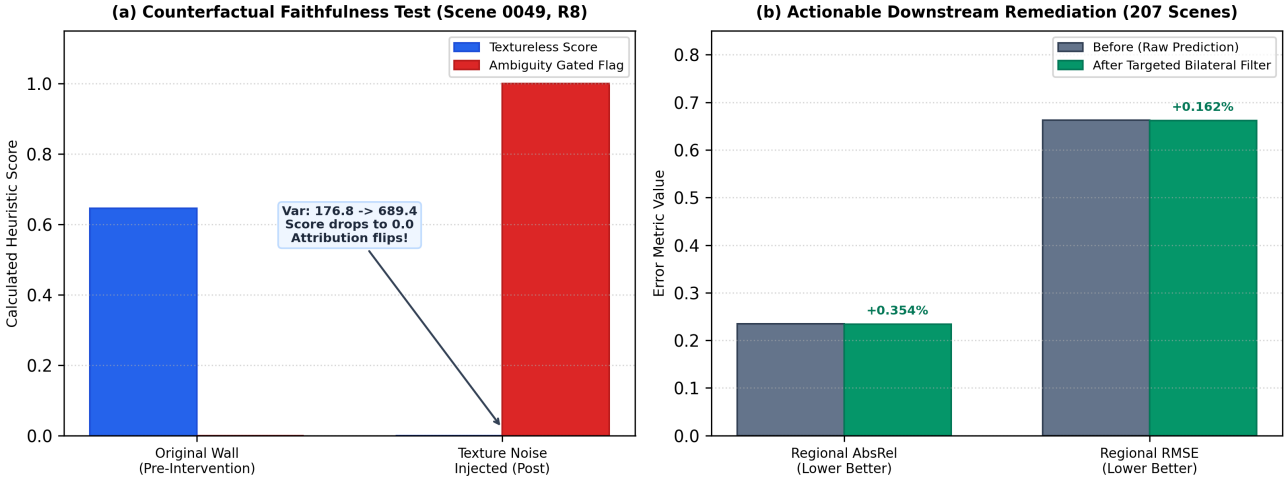


Fig. 5. Counterfactual Faithfulness and Downstream Remediation. (a) Texture noise injection into Scene 0049 Region 8 collapses textureless score, cleanly flipping attribution to ambiguous. (b) Targeted bilateral filtering on 207 scenes yields +0.354% regional AbsRel reduction.

7. Discussion & Engineering Remediation

Our diagnostic findings provide actionable architectural recommendations: (1) add edge-aware depth gradient losses ($L_{grad} = ||\nabla \hat{D} - \nabla D_{gt}||$) to penalize boundary bleeding; (2) apply synthetic frequency injection (CutBlur, Perlin noise) to break over-reliance on surface speckles; and (3) supervise with non-Lambertian synthetic rendering passes to discard specular virtual depths.

8. Limitations & Threats to Validity

We disclose four key limitations: (1) Correlation vs. Causality: VEO measures statistical correlates of failures; unconditional causal proof requires physical interventions; (2) Sensor Artifacts: Kinect infrared absorption holes create valid-mask boundaries where gradients may exhibit slight distortion; (3) Rater Subjectivity: Human agreement ($\kappa = 0.348$) establishes an upper bound on manual consensus; and (4) Conservative Gating: 39.3% of regions are flagged as ambiguous due to narrow margins (< 0.10), prioritizing scientific caution over forced guesses.

9. Conclusion & Future Work

The Vision Error Observatory provides a transparent, model-agnostic failure diagnostic framework for monocular depth estimation. By moving beyond aggregate scalar metrics, VEO enables researchers to identify why models fail, expose architectural trade-offs, and implement targeted post-processing and training remedies.

Declarations

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